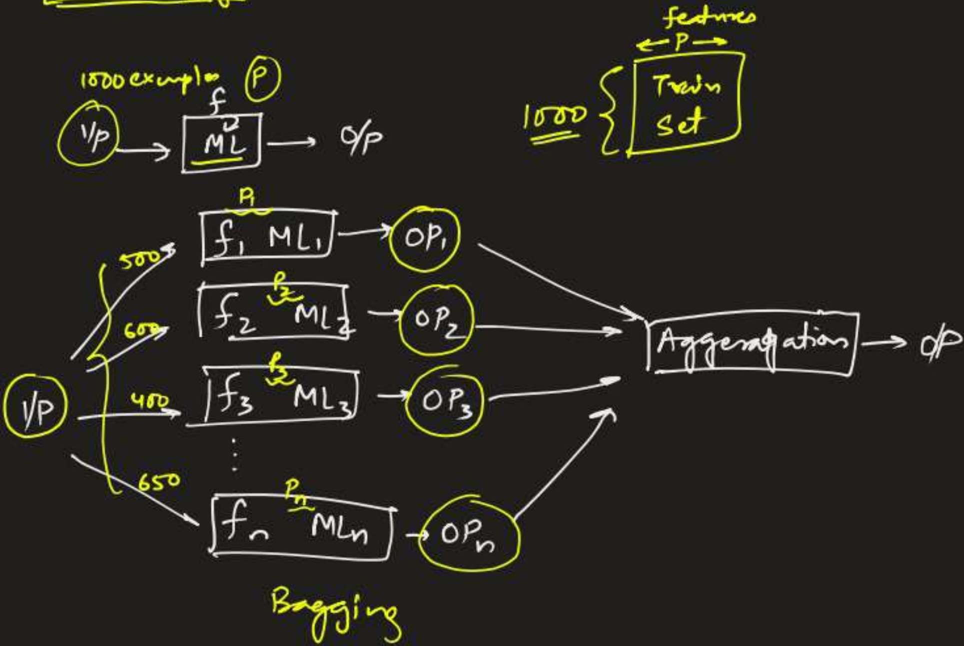
Ensembling



Ensembling techniques are powerful methods in maching learning that combines multiple ML models to improve predictive performance.

By aggregating from various models, ensembling reduces overfitting, improves accuracy and enhances robustness compared to individual models.

Ensembling Techniques

- **Bagging** (Bootstrap Aggregating) - Multiple instances of the same learning algorithm are trained on different subsets of training data.
- **Boosting** - Models are trained sequentially. Each subsequent model focuses on correcting the errors made by previous models.

Bagging

Suppose a set of N examples (D) are given for training each containing p dimensional input vectors.

At each iteration i, a training set D_i of n tuples is selected via row sampling with replacement method from D (ie bootstrap)

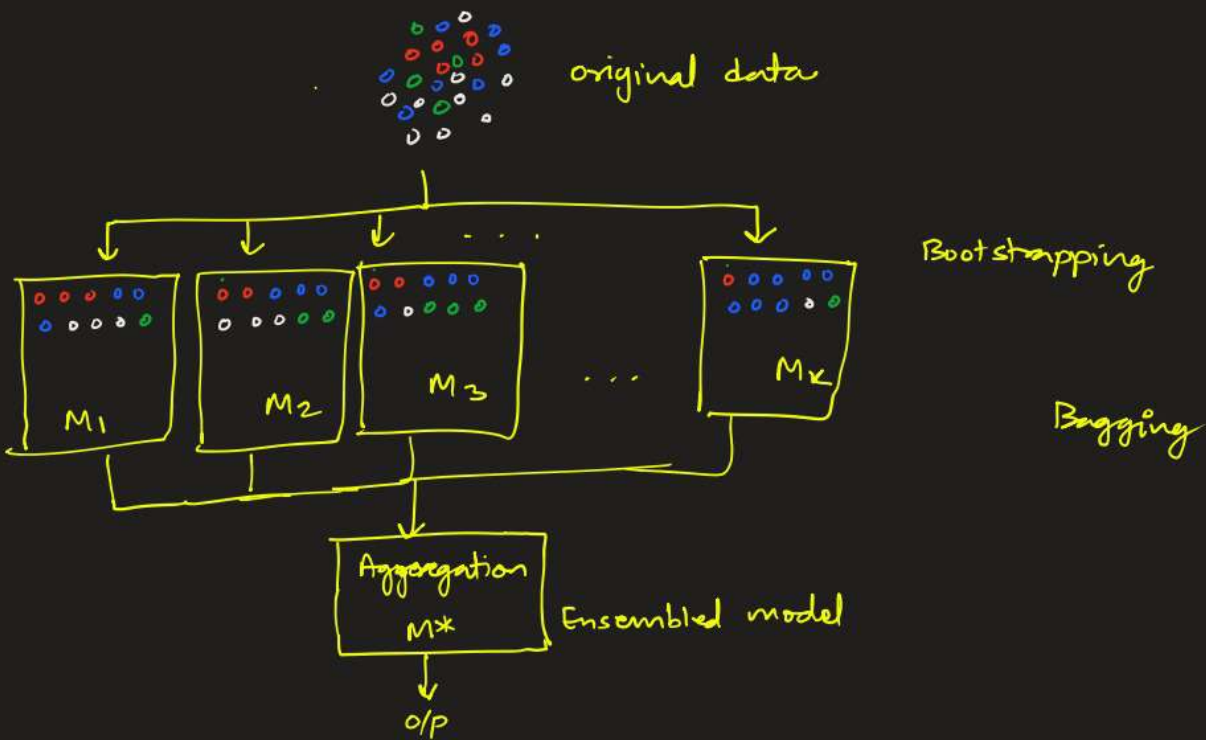
Then a classifier model M_i is learned for each training set D_i.

Each classifier returns its class prediction. The bagged classifier M^* counts the votes & assigns the class with most votes to an unknown sample.

* Bootstrap sampling :- Random selection with replacement

* Random Feature selection :- A random subset of features is considered for training models.

* Aggregation :- Majority voting (classification)
  Average the predictions (Regression)



Bootstrapping:

Bootstrapping is a statistical resampling technique that involves repeatedly sampling with replacement from a dataset.

- From the original set of size N, multiple bootstrap samples of size n $(= N)$ generally are created. Each point in each sample is drawn from the initial dataset randomly, allowing for the possibility of selecting the same instance multiple times.

Training set

ID	Feature 1	Feature 2	class
1	A	1	Yes
2	B	2	No
3	A	2	Yes
4	B	1	Yes
5	A	3	No

$N = 5$

Not selected in bootstrap

Out-of-bag

Bootstrap →

In bag

ID	Feature1	Feature2	Class
3	A	2	Yes
1	A	1	Yes
4	B	1	Yes
1	A	1	Yes
3	A	2	Yes

$n = 5$

→ $\boxed{M1}$ ...

→ ... → $\boxed{M2}$ ...

⋮

→ ... → $\boxed{M_K}$ ...

Out-of-bag fraction:- $\dfrac{\text{\# data points not in bag in a bootstrap}}{\text{\# data points in the dataset}}$

In bagging, the out-of-bag data points can be used as the validation set for corresponding model.

Bagging Fraction :- $\dfrac{\text{\# unique data points in the bag}}{\text{\# data points in the dataset}}$

- Bagging Fraction ~ 63.2% if $n = N$ and N is large

Proof: $P(\text{rejected}) = \left(1 - \frac{1}{N}\right)^n = \left(1 - \frac{1}{N}\right)^N$.

if N is large $\lim_{N \to \infty} P(\text{rejected}) = \lim_{N \to \infty} \left(1 - \frac{1}{N}\right)^N = \frac{1}{e}$

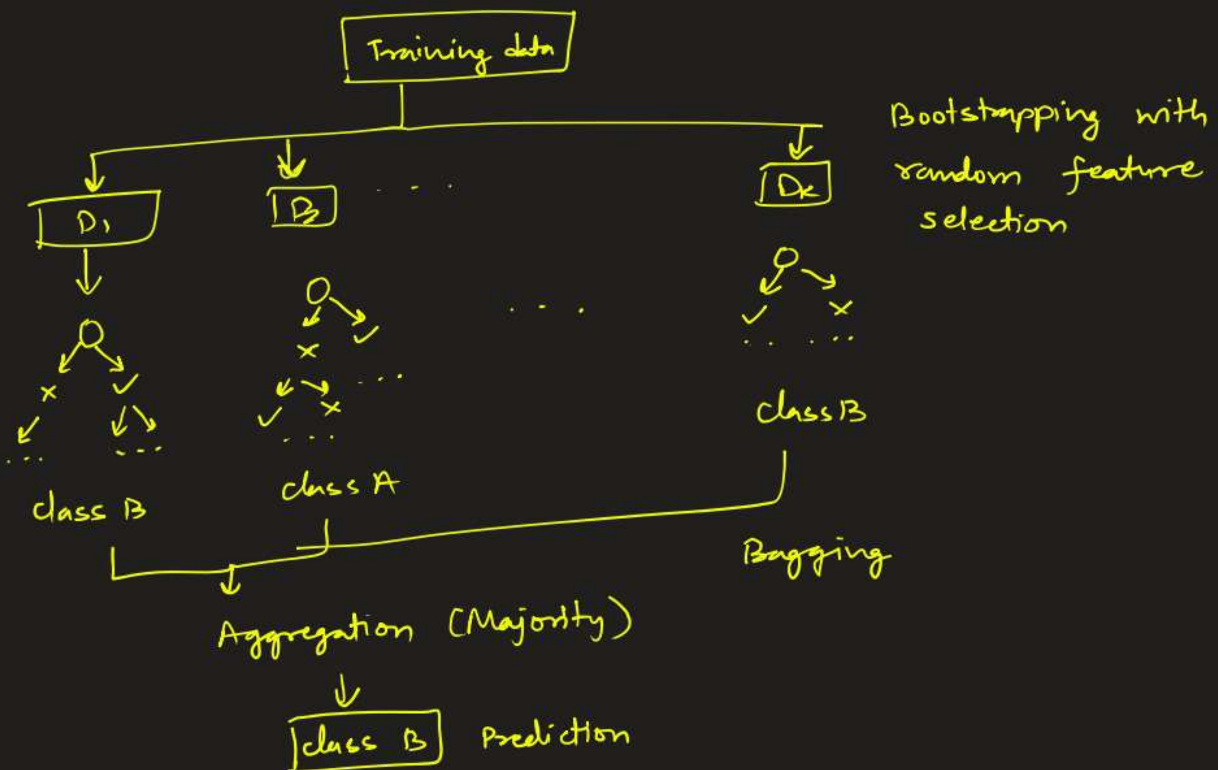
≈ 0.3679

$\therefore$ P that a particular is NOT selected $\approx 36.79\%$

$\therefore$ P that a particular element is selected ~ 63.2%.

Random Forest

↳ is an extension of bagging that enhances the decision tree algorithm. We build multiple decision trees on bootstrapped samples with random feature selection and then aggregate the results produced by these decision trees.



Advantages

- High Accuracy
- Robustness
- Reduced Overfitting
- Feature Importance
- versatile

Disadvantages

- Complexity
- Memory Consumption
- Longer training time
- Bias in Predictions